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The Road Towards Service-Integrated Offerings for Product-Centred Manufacturing Firms

A Case Study within Husqvarna

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Abstract

In recent years there has been an increase in servitisation amongst manufacturing firms. Change could be inevitable to stay competitive in this fierce innovative environment and is therefore an ongoing topic that needs to be addressed. This impacts how the firms need to adapt and position themselves in order to survive, though the path to become service-integrated is not apparent. The purpose of this master's thesis is to perform a case study of Husqvarna, a world-leading manufacturing firm within forestry, gardening and construction, known for its premium quality and brand, in order to discover what is needed both to become product-service integrated, as well as to select and launch such offerings. This thesis investigates these areas and offers a thorough framework that may be used as a compass by manufacturing firms looking to undergo long-term change. However, there are many challenges in becoming a service-integrated firm and meeting the requirements needed. An analysis of gaps within this research field was conducted which led to interest in impacting shifts, uncertain criteria and maturity for product launch. The framework sheds light on the crucial elements involved in the servitisation journey and was developed through a thorough synthesis of literature and results from

data analysis. The insights were obtained through semi-structured interviews and formal meetings with experts in the field and relevant personnel within the firm, bridging the gap between theory and reality. It consists of three pillars covering transformation, concept screening, and launch. These pillars include clear suggestions on steps towards becoming a product-service-focused firm with successfully released offerings. Firstly, it has been discovered that to become a product-service-focused firm there needs to be openness for acquisition of new mindsets, resources and procedures that benefits the service integration. Secondly, it is important to choose the right ideas and concepts to realise, by making sure they are technically feasible and create value for the customer, backed up with solid collaboration within and outside the firm. Thirdly, product-service offerings have proven to be different from physical products in their required maturity to launch, meaning that they can be released while still incomplete with continued development thereafter. There is however a certain level of completeness that needs to be fulfilled, such as establishing a minimum viable product and continuous feedback loop for the offering during the whole product development process.

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KTH Industriell teknik och management

Vägen mot tjänsteintegrerade erbjudanden för produktcentrerade tillverkningsföretag

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Sammanfattning

De senaste åren har visat på en ökning av intresse för tjänstefiering inom produkttillverkningsföretag. Förändring är med stor sannolikhet ett måste för att fortsätta vara konkurrenskraftig inom denna bransch och är därför ett område som behövs undersökas. Detta tvingar företag att anpassa sig och positionera sig följaktligen för att fortsätta vara relevanta, dock så är vägen till att bli mer tjänsteintegrerad inte självklar. Syftet med denna masteruppsats är att genomföra en fallstudie på Husqvarna, ett världsledande produkttillverkningsföretag inom skog, trädgård och konstruktion, välkända för deras högkvalitativa märke, för att kunna undersöka vad som behövs för att bli mer produkt-tjänstinriktat, samt hur man idéer utses och när de ska lanseras. Denna uppsats undersöker detta område och erbjuder ett grundligt ramverk för att navigera PTF då de genomgår denna förändring. Dock finns det ett flertal utmaningar och krav att uppnå för att bli mer tjänsteintegrerat som företag. En analys av möjliga luckor inom forskning genomfördes för att hitta intressanta områden som globala förändringar, krav på produkter som behövs, samt även mognadsgraden för lansering. Det presenterade ramverket upplyser viktiga element som behövs inom tjänstefieringsprocessen, vilket var

utvecklat utifrån en noggrann sammanställning av litteratur och insikter. Insikterna var framtagna från semi-strukturerade intervjuer och formella möten med experter och relevant personal inom företaget, för att skapa en övergång mellan teori och verklighet. Ramverket är uppbyggd av tre pelare som täcker transformation, konceptscreening och lansering. Dessa pelare inkluderar förslag på de steg som behövs för att uppnå en lyckad tjänstefiering. För det första behövs en öppenhet och förvärv av nya tankesätt, resurser och tillvägagångssätt mot tjänsteinriktning. För det andra är det viktigt att välja rätt idéer och koncept för ett projekt. Där några av kraven är att se till att de är teknologiskt möjliga samt att de skapar värde för kunden, byggt på ett samarbete inom och utanför företaget. För det tredje visar det att produkttjänsterbjudanden har en annan mognadsgrad av lansering än fysiska produkter, där en lansering av dessa erbjudanden kan lanseras ofullständiga. Dock måste minimala krav uppnås, där minsta livskraftiga produkt och kontinuerlig feedbackloop är avgörande för att bibehålla ett erbjudande under hela produktutvecklingsprocessen.

Foreword

This master's thesis has been a collaborative work to present the combined results from literature and empirical data. This thesis delves into a topic that has piqued our attention and devotion during the duration of writing our master's thesis. Said topic compelled us to examine a fitting way to go from a product manufacturing firm to a service-integrated one. We hope for our discoveries to benefit both the academic community and the real world of product-service offerings.

We had the honour of getting priceless advice from our outstanding advisers, Jennie Björk at KTH, as well as Gustaf Lagercrantz and Johan Simonsson at Husqvarna, while we worked to develop our thesis. Their knowledge, guidance, and constant support have been crucial in determining the calibre and focus of our thesis, as well as making sure we upheld the right quality.

Husqvarna, the firm with whom we had the privilege to work and conduct interviews with, deserves our special recognition, a special thanks to all informants who participated. Our study has greatly benefited from their openness to share their knowledge and experiences. In addition we want to thank KTH for giving us this connection and valuable learning opportunity.

We have worked in close proximity with our fellow colleague Robert Bunea who we have exchanged and brainstormed ideas, as well as set up interview guides and conducted interviews together with.

We want to thank everyone who has helped us in this endeavour once more.

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1. Introduction

Although the path to getting there is unclear, many product manufacturing companies are going through servitisation to become more product-service-focused. The challenge does not only lie within the transformation itself but also with the offerings that are being chosen for development as well as how and when they are being launched.

1.1 Background

Husqvarna is primarily a manufacturing firm that currently wishes to expand its offerings within services to also offer a blend of products and services. Currently being a traditional manufacturing firm and trying to develop services does present some challenges; for example, understanding how the integration of a service affects the development of a physical product.

The transition of a traditional manufacturing firm towards more product-service offerings is of great interest for firms of this type, mainly because of pressure from different forces, such as more varied and unique customer needs, increased market competition on a global scale, and ongoing sustainability issues (Baines et al., 2009; Lerch & Gotsch, 2015). These offerings are highly integrated between product and service to increase customisation and offer more value towards customers. Additionally, a successful firm transition paired with product-service offerings has been shown to lead to both economic and environmental success (Mont, 2002). However, undergoing such a transition has proven to be a challenging task for most traditional manufacturing firms because it requires major changes at an organisational and individual level (Lindahl et al., 2014). Despite there being examples of successful transitions, scholars have been researching what has made some firms succeed and been able to develop processes or models that practitioners may follow (Fargnoli & Haber, 2017a). Regardless of the large amounts of models and processes from research, there has not been a general consensus

regarding concrete steps for firms to follow to achieve a successful transition (Beuren, Ferreira & Miguel, 2013; Fargnoli & Haber, 2017b).

This thesis will contribute to this field of study by researching new possibilities for traditional manufacturing firms to take while undergoing this extensive transforming phenomena within the organisation; by also taking into account the turbulent growth of undergoing shifts such as digitalisation and artificial intelligence (AI).

1.2 Purpose and Definition

The purpose of this project is to explore the main challenges to consider when transitioning from a product-centred firm to a product-service integrated one. This will require grasping an understanding of how current processes work both internally and externally of the firm. Additionally investigating how the relationship between other interests such as customers and third-parties work, in correlation to product-service offerings collaborations. Another part of this project is to introduce a framework to follow and to guide when transitioning from a product-centred firm to a product-service integrated one; while considering important challenges such as how to set up relevant criteria or what is required for the launch of a product-service offering. These formulated research questions were established after a thorough theoretical background was performed, and an investigation of what gaps were present in this research area:

RQ1. What are the main challenges to consider when transitioning from a product-centred firm to a product-service integrated one?

RQ2. What criteria are required for product-service ideas to proceed in the innovation process?

RQ3. What is the right maturity and relevance for the launch of a product-service offering?

2. Theoretical background

This chapter will cover the theoretical background consisting of data collected from articles covering relevant areas of interest. The purpose of this chapter is to give insight into how thoroughly researched this area presently is. The theoretical background will also give an understanding of how similar firms have been successful or unsuccessful in previous attempts to transition into becoming more product-service-oriented. This will include a thorough comprehension of how a traditional product development process is compared to a product-service one, from the ideation to the launch of the respective offering. Also, studying underlying reasons for why certain decisions were made and what could have been performed differently, examining if and how external factors, such as technological aspects impact the firm's direction. This will provide useful knowledge when analysing the empirical data of the studied firm's situation, in order to evaluate a guidance to avoid possible pitfalls ahead.

2.1 The shift in the manufacturing industry

To find what the main challenges to consider when transitioning from a product-centred firm to a product-service integrated one are, it is important to inspect what this shift actually entails, how it is defined and lastly what challenges have already been discovered through the transition of previous firms in the past.

The constant undergoing shifts are forcing companies to rethink their strategies and sustainability in the market in order to stay relevant and competitive. Shifts in the last decades could be global effects such as digitalisation or the recent radical development of AI impacting how firms are managing and organising their innovation processes (Johnson et al., 2022). However, not all shifts arise due to innovation, as some shifts occur unexpectedly and rapidly, such as the pandemic of COVID-19 or climate change. No matter what type of shift, whether it is driven by technological, societal, or regulatory demands, these shifts tend to affect customer behaviours which force companies to adapt accordingly (Am et al., 2020). For manufacturing firms, which tend to heavily rely on

their conservative product-oriented business models, sale of product, have received a lot of headwind due to these shifts, especially technological shifts (Engwall & Tongur, 2018). However, in terms of headwinds because of shifts, one could also perceive this as an era of opportunities where one path or solution for manufacturing firms to stay competitive in these changing environments is to integrate or transition themselves into a Product-Service System. This process is often referred to as servitisation.

2.2 Servitisation

The transition of going from pure manufacturing to Product-Service Systems (PSS), and offering services together with products has become increasingly popular among manufacturing firms as a way to find new value delivery and profit. This transition is known as servitisation. (Martinez et al. 2010; Isaksson, Larsson, & Öhrwall Rönnbäck, 2009; Rapaccini, 2015; Vandermerwe & Rada, 1988)

Servitisation occurs on a predefined path and is divided into different stages, ending in PSS. This transformation path is shaped based on the characteristics of the manufacturing firm, with the influence of digitalisation and servitisation (Lerch & Gotsch, 2015). Some of these stages require the firm to use services as a main aspect for competitive advantage while other stages think of services as add-ons or as tools to sell the products (Martinez et al., 2010). There are four stages, where the first stage is product-related services such as repair and maintenance. The second stage focuses on digitalising the product and gathering product data from a distance. The third stage is pure, but separate digital services that enhance the product offering. Last is the fourth stage which represents PSS (Lerch & Gotsch, 2015). This highly suggests that to even reach the fourth and final stage, a completion of all previous stages must be fulfilled. Also, the importance of understanding what PSS is, its definition and what it can provide for the firm is a vital factor before even considering adopting it.

2.3 Product-service system (PSS)

PSS are less likely and harder to be copied by competitors compared to products on their own (Dickson, 1992). In addition to that, PSS can be used as a new way to fulfil customer needs, it can also reduce life cycle costs and increase revenue (Isaksson, Larsson, & Öhrwall Rönnbäck, 2009).

However, adapting to PSS can become a significant investment early on because of the need for the firm to acquire capabilities and resources that it currently does not possess (Beuren, Gomes Ferreira, & Cauchick Miguel, 2013; Adrodegari, Saccani, & Kowalkowski, 2016; Rese, & Gesing, 2013; Parida et al., 2014). Each stage will however eventually reach a stable point where the firm can minimise its innovation investments as long as the service offering stays the same. At this point, funding can go towards the sales of existing PSS. When factors disturb the stability, the firm will need to continue on through the transformation path until the next stable point. Eventually reaching a stable point with PSS (Lerch & Gotsch, 2015).

Definitions of PSS

In terms of definitions of PSS, which has arguably been on a rise the last few decades and a relatively new term to understand. As of today, there is a definitive consensus amongst researchers that the existence of three categories or different PSS types of business models are to be referred to (Barravecchia et al., 2021). The three categories which were presented by Tukker (2004) could be seen as the foundation block of PSS.

1. Product-oriented services.
2. Use-oriented services.
3. Result-oriented services.

The product-oriented services category is when the business model is still focused on the traditionally known linear sale process, sale of product, while adding extra services coincided. Within the product-oriented services category, the first sub-category is

product-related services which according to Tukker (2004) is to not only sell the product itself but also provide relevant and important services during the use of the product. This could for example be maintenance contracts, financing schemes or supply of consumables, and a take-back agreement at the end of the life cycle. The second sub-category is, *advice and consultancy*, which means alongside with selling the product, providing essential advice to the consumer, for example, on what the most efficient usage is (Tukker, 2004).

Use-oriented services are still using the respective product as a core of its services, however, the business model is not focused on the actual sale of the products. This category is when the ownership remains with the provider during the entire process by offering it in a different way through renting and leasing for example. With *product lease* as a sub-category, the maintenance, repair and control is overseen by the provider, where an unlimited access is available for the consumer. The next is *Product renting or sharing* where the maintenance, repair and control is the same as leasing, however, the availability is not unlimited as there is a possibility that the product may be used by other consumers during downtime. This is similar to the last sub-category which is *Product pooling* which instead is the definition of using the product simultaneously (Tukker, 2004).

Result-oriented services is where an agreement between the provider and the consumer is established to reach a desired result for both parties. The main part of this business model is that there are no fixed products involved to reach the agreed result. The first sub-category being *Activity management/outsourcing* which refer to allow a part of the organisation being outsourced to a third party (Tukker, 2004). Outsourcing can be performed because of various reasons, for example, to reduce costs, reduce risks or lacking expertise within a field (Rehman et al., 2018). The second sub-category is *Pay per service unit* which is when consumers do not purchase the product but the output instead. A prime example could be the pay-per-print formula, where the consumer only pays for the printed paper rather than the whole machine which is being monitored and maintained by the provider. The last sub-category is *functional result* and is primarily focused on what should be achieved. An example could be to provide a pleasant climate

in an office, however, the approach is entirely up to the provider to reach the desired result (Tukker, 2004).

The definitions of servitisation and PSS provide important insights on how to position as a firm and in which direction to continue developing its services. As well as guiding this thesis in what to recommend and conclude. However, to achieve a successful integration of PSS, a firm would certainly need to meet a great deal of challenges.

Challenges for successful integration of PSS

Although PSS is one of many solutions a manufacturing firm can strive for or take to achieve great competitiveness, a successful integration comes with many challenges. There are mainly three identified large challenges firms have to tackle according to Vasantha et al. (2011) in order to develop a sustainable and successful PSS;

1. Ontology for PSS.
2. System modelling techniques in PSS design.
3. Sustainability.

The first barrier is ontology for PSS which refers to all terminology and relations among them (Vasantha et al., 2011). Spreading knowledge throughout the organisation can be seen as a time-consuming engagement from the management to achieve an efficient knowledge stream, however, nevertheless argued to be essential for enhanced innovation and performance (López-Nicolás & Meroño-Cerdán, 2011).

System modelling techniques in PSS design will not only require time and personnel but also additional knowledge to break down and create new systems as well as develop and adapt old ones. This has proven to be challenging for dominant manufacturing firms due to their historical records which are dependent on an already-developed system (Wallin, Parida & Isaksson, 2015).

Sustainability involves environmental, societal and economic aspects to consider, which is a major hurdle when transitioning to a large-scale organisation as their dimensions intertwine, and affect a great deal of factors when changed (Beuren, Gomes Ferreira & Cauchick Miguel, 2013).

To further build upon these challenges, Martinez et al. (2010) have identified five additional possible challenges that manufacturing firms have to tackle when integrating systems.

1. Strategic alignment.
2. Embedded product-service culture.
3. Delivery of integrated offering.
4. Internal processes and capabilities.
5. Supplier relationships.

Similarly to the ontology for PSS, the alignment of strategy requires a shared language, with the addition of a shared mindset for service, both within the firm and with the customers. This is so that the firm can be more aligned in different parts of the firm as well as with the customers. To achieve this, transparency is key (Baines, Lightfoot, & Kay, 2009). There is a need to change from the traditional mindset of developing a product and thereafter developing a service as an add-on, to developing a solution as a whole. (Isaksson, Larsson & Öhrwall Rönnbäck, 2009; Story et al., 2017).

When it comes to embedding the product-service culture, the manufacturing firm will need to integrate the product-service culture into its own culture. In order to satisfy the customer's expectations, and be more customer-centric (Story et al., 2017). However, a strong manufacturing culture can increase the difficulty of integrating a new culture, as well as new strategic options that oppose the previous culture (Schein, 1990). Integrating services into the culture will move the firm's value creation from being transaction-based to relationship-based (Vargo, Maglio & Akaka, 2008).

The delivery of an integrated offering requires the firm to focus on the offering as a whole. A common mistake that usually happens when new products are released, is that the firm only focuses on the product and not the whole offering (Martinez et al., 2010).

Internal processes and capabilities represent the need to get new capabilities. Knowledge sharing and understanding make it easier to measure processes and performance. However, oftentimes firms are working excessively in silos rather than collaborating to create the best customer experience (Martinez et al., 2010). It is important for firms to acquire new competencies to transition to PSS, where most of these competencies revolve around the customer. More competencies are also needed because the project scope has been expanded (Isaksson, Larsson, & Öhrwall Rönnbäck, 2009).

Finally, supplier relationships are strongly needed for integrated offerings, where cooperation between the provider and the supporting network is highly important. Limited information sharing between supplier and provider can lead to unwanted effects. If parts of the offering seem as separate, suppliers may feel biassed towards what they help invent, pushing away the other part (Martinez et al., 2010).

The supplier relationship is certainly important to consider, but this is not the only relationship that needs to be focused on when going through servitisation. Another vital relationship that is extremely valuable in this transformation is that with the customer.

Customer involvement within PSS

Customers are already important in traditional product development. However, to offer PSS, the customer is essential in development, and there is a need for even more customer participation throughout the whole development process. The offering needs to be designed for and together with the customer, meaning the customer needs to partake in giving data regarding their behaviours, preferences and requirements for the design of the product (Isaksson, Larsson, & Öhrwall Rönnbäck, 2009). Constant contact with the customer is necessary to increase the customer's understanding of new offerings and

avoid misunderstandings. On the other hand, a lack of customer contact could raise the risk of misunderstandings occurring regarding customer needs, hence not being able to fulfil them. This is especially important in the innovation process (Mahr, Lievens & Blazevic, 2013).

Unlike manufacturers, dealers have a direct connection with the customer. Manufacturers should therefore include dealers in the product development process to make use of this source of customer information regarding customer demands and concerns (Pitta & Franzak, 1996). Another way to connect with the customer is through digital tools and channels. Compared to the low cost of digital channels, firms that rely on in-person contact with the customer may realise its high cost and susceptibility to error, and therefore fully give up on customer contact thereafter (Mahr, Lievens & Blazevic, 2013; Thomke & Von Hippel, 2002).

There are without a doubt a lot of challenges to take into account when becoming more service-integrated, for example, the changes that need to happen within the organisation, the new acquisition of resources and capabilities, as well as the relationships needed to uphold the right standard. However, a service-integrated firm is still only as good as the offerings it provides, and therefore, it is important to also understand what specific criteria are needed for products and product-service offerings to proceed in the innovation process to meet customer needs and expectations.

2.4 Criteria for advancing to the next phase in the innovation process

The innovation process can certainly be unique for each and every organisation, where some use methods and models to aid their journey towards their goal, while others might use an open-ended process instead, however, both yield the same result, where some end up as successes and a great deal end up as failures. To understand what criteria are required when screening for the right product-service ideas to proceed in the innovation

process, it is important in this thesis to look into what criteria are currently used to narrow down to the most successful ideas. However, limited information could be found on the screening of product-service offerings, as well as for services alone.

It is argued that the more an organisation understands the term innovation, the higher its success rate is (Kahn, 2018). Though organisations use different strategies to reach the final stage of a process, the market launch of a product and a service, the journey there often includes several stages or phases. The interesting key question remains as to what different factors determine or what criteria need to be met in order for an idea to proceed to the next stage.

Traditional product development criteria

Balachandra (1984) sees strict and clear criteria in two sets of significance for success and failure. The first set consists of four criteria that must be fulfilled, or else the product risks failing.

1. Having technology required for the product needs to be available.
2. Access to raw materials is required or market potential could be reduced as well as an increase in production cost.
3. A market that is expected to remain for a long time, or else the need for the product could risk dissipating.
4. Firms need to avoid the use of anything that is at risk of being regulated by the government and therefore terminated.

If any of these criteria are not fulfilled, the project should be put on hold until they are fulfilled. The second set of criteria are important but not as significant on their own to the success and failure of the product. Some of these are the chance for commercial success, personnel support, amount of end users, competition and profitability. If too few of these criteria are fulfilled, the project should be terminated. On the contrary, if the majority are fulfilled, it is expected to succeed (Balachandra, 1984).

Cooper & Brentani (1984) brings up eleven factors in the screening stages, where four of them were seen as most critical;

1. Financial potential, expecting both to profit and growth.
2. Corporate synergy between sales, the firm, and the customers.
3. Technological and Production Synergy, meaning the firm has access to the resources and facilities required.
4. Product Differential Advantage by being high quality, novel and having the potential to be a leader in the market.

For traditional product development processes, early on, during the idea and concept screenings, technical feasibility and market potential are important evaluation criteria. During the business analysis, this will shift towards sales objectives and sales in units. Lastly in the product testing, criteria such as product performance and quality stand out as what the product gets evaluated on (Hart, et al., 2003). Frederiksen & Knudsen (2017) agrees with Hart et al. (2003) about the importance of market potential. They also bring up the novelty of the product which Hart et al. (2003) does mark as a factor, though less significant than previously mentioned ones.

Service-focused product development criteria

In a service-focused innovation process, not only do customers play a vital part in the early phases to provide ideas and demands, they also act as a guide throughout the whole process by testing and ending up purchasing the products, thereafter relaying feedback which then can be turned into ideas. A crucial objective throughout the whole process should be the balance between effectiveness and efficiency. During the early stages of the innovation process, effectiveness should be prioritised which is a focus on high quality and appropriate services at a reasonable price. Right before market launch, efficiency will be the larger factor as the providing offer should in the end be considered to pay off for the provider (Dörner, Gassmann & Gebauer, 2011).

Dörner, Gassmann, & Gebauer (2011) have conducted a 10-year study with a focus on how to achieve a successful service-innovation process within the organisation. They divide the innovation process into three phases, where the first one is definition, which then is

followed by development and lastly ends by market launch. Important criteria to consider in the definition phase, should be focused on investigating and sorting ideas through customer demands. In the development phase, the focus should shift to tool-building, modularisation and ensure a sustainable development process before reaching the final phase, market launch. See Figure 1 for a detailed summarisation of a service-focused innovative process.

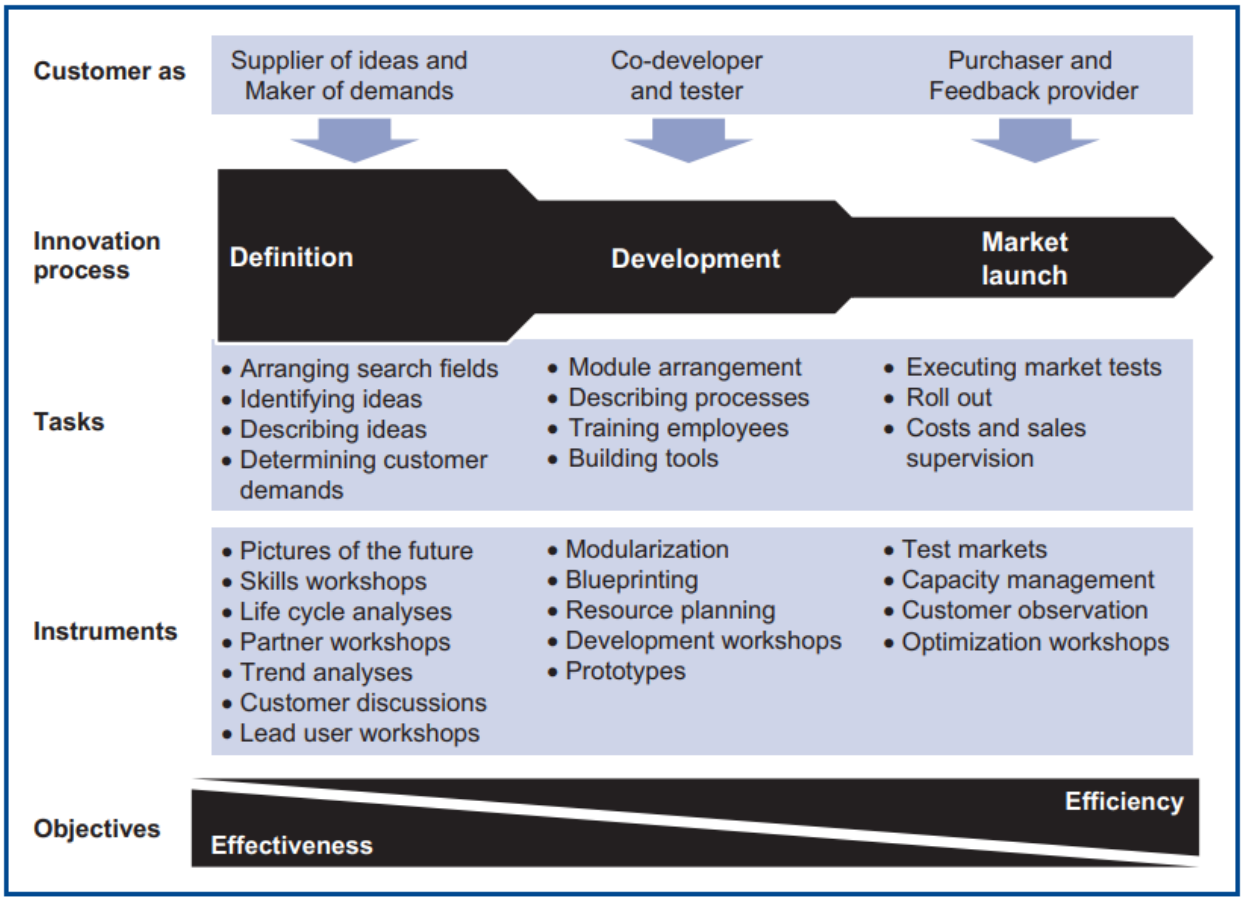


Figure 1. Innovation process for services (Dörner, Gassmann & Gebauer, 2011).

2.5 The right time to launch a new product-service offering

The million-dollar question many companies ask is when the right time to launch a new product on the market is. This could certainly be argued as one of the most vital key factors to becoming a leading or successful organisation in a fierce innovative

environment (Tholkea, Hultinkaa & Robbenbb, 2001). Previous studies show that establishing what type of firm it is in question is one of the core factors to consider in order to determine a relevant strategy for product launch. The type mentioned is determined by the estimated market size and product development capabilities of the firm, which thereafter determines the key question of when a product is launched (Bayus, Jain, & Rao, 1997). As large and technologically leading firms are usually the first ones on the market, the smaller ones would later follow. However, although Bayus, is Jain & Rao (1997) determined a correlation between market size and market launch, some extensive projects that were launched as first-movers by large and leading firms did not always land well amongst customers. Making the key question an even greater complex area to understand, that nothing is truly set in stone as new experiences and innovations do not always definitely equal success. To understand the right maturity for a product-service offering to launch in this thesis, it is important to look into what the literature deems to make or break a successful product-service launch.

Key factors for a successful product-service launch

As already stated, the right time for a successful product launch is not a fixed formula and has proven to be even more difficult for newly introduced services. However, despite the fact that it is difficult, through research and analysis, some key areas successful launches have in common that could be distinguished and categorised into;

1. Well-established digitalisation within the organisation (Chester Goduscheit & Faullant, 2018; Sklyar et al., 2019).
2. The level of understanding of competitors (Bayus, Jain, & Rao, 1997; Grawe, Chen & Daugherty, 2009).
3. Identified potential risks in trade-offs (Bayus, Jain, & Rao, 1997; Ettlie & Rosenthal, 2012).
4. Customer involvement (Dörner, Gassmann, & Gebauer; 2011; Ettlie & Rosenthal, 2012; Grawe, Chen & Daugherty, 2009; Miles 2010).
5. Managerial incentivisation (Dörner, Gassmann, & Gebauer, 2011; Ettlie & Rosenthal (2012); Miles, 2010).

The first category is understanding the need for digitalisation within the organisation to enable a service innovative environment. This highlights the fact that in order to even properly launch any type of service as a manufacturing firm of origin, an understanding of the firm's capability and capacity is extremely important, to be able to visualise one's digital boundaries (Sklyar et al., 2019). Managerial challenges will also need to direct their focus on understanding the utilisation of data-driven innovation. Understanding the value of data and what the organisation can obtain by implementing these services is an important underlying factor to consider, rather than primarily providing a service while not considering what can be gained from it (Chester Goduscheit & Faullant, 2018).
The second category is how well a firm understands its competitors. This cannot only provide an understanding of what others are good at but also what can be needed within the organisation as well by analysing strengths and weaknesses outside and inside the firm. This is required to not launch a product too early or too late (Bayus, Jain, & Rao, 1997; Chen & Daugherty, 2009). Building on the second category, the third category is the level of understanding of what an organisation can potentially afford and risk trade-offs in order to perfectly match time-to-market, product performance and development costs. Not all projects will end in successes, therefore a vital step to grow as a firm is allowing failures and mistakes in order to create an inviting innovation environment (Ettlie & Rosenthal, 2012). The firm should also be able to answer important questions such as if a need for a shorter lead time is the goal, then what can be sacrificed to achieve the respective goal (Bayus, Jain, & Rao, 1997).

The fourth category to achieve great PSS is customer involvement which also plays a vital role in the progress towards a successful product-service launch. The key is to establish a direct customer relationship with the whole chain, from end-customer to manufacturer in order to develop an in-depth understanding of customer needs (Miles, 2010). Having this direct link of relationship allows the firm to gather information from customers that do not express feedback verbally, which is the prime example of how customers do not often do what they say they do, but rather what they think they are doing. Lastly, the importance of having managerial incentivisation to allow for successful new service

development should be considered a core subject within the firm, to encourage and inspire service innovations to grow (Ettlie & Rosenthal, 2012; Miles, 2010). This is a step many manufacturing firms struggle with as they usually tend to invest small amounts in service development because of the uncertainty of wandering into the unknown (Dörner, Gassmann, & Gebauer, 2011).

Another rising uncertainty to consider is the growth and rapid development of groundbreaking digitalisation and AI solutions, which is becoming more prominent as of today. Their dominance is a subject hard to ignore, if not impossible in today's society which adds an additional complexity to when a firm should launch a new product, a complexity that has dramatically increased over the last few decades (Soni et al., 2020).

Impacts of digitalisation on new product launches

Digitalisation is rooted in reprogramming and converting analogue inputs into digital data (Yoo et al., 2012; Dufva and Dufva, 2018). In this new area of digitalisation, a new approach of not dividing the process and the result emerges. By releasing an incomplete version of the offering, it is still open to adaptability and change in the preferences and feedback of the users (Garud, Jain & Tuertscher, 2008). The firm would need to revise and build upon its new offering based on customer feedback after the initial launch (Lehmann & Recker, 2022). Optionally, before a service is officially launched, the firm can conduct pilot testing of the service. Where a limited group of actual customers will get to try the offering under observation and then provide feedback. This is to see if the offering is stable enough for launch and if changes could be made (Rapaccini et al., 2013).

The firm could release a soft launch, here meaning releasing the product gradually. Building upon and replacing part by part in stages iteratively (Couldwell, 2019). This initial launch should deliver on the main values of the offering that actually provides benefit to the customer. The firm needs to make sure that the first basic version of the product can still solve the problems promised by the offering and therefore be the main selling point of the offering. It is important to note that for an early launch, mainly aimed

at early adopters, quality overtakes quantity. A few working functions will be given, not to disappoint the users or overpromise. New functions can then be added at a later time. Slowly building quality from the inside out (Ries, 2011). To make sure that what is offered is also wanted, the firm needs to create a minimum viable product (MVP). An MVP is a product with the fewest possible features giving the greatest amount of value (Maurya, 2012). This will help the firm get as much information as possible about the user for the least amount of money and work, also guaranteeing a faster release (Lenarduzzi and Taibi, 2016). There is no underlying structure for the balancing of an MVP, and it has to instead be based on judgement. In most cases the simpler the better (Ries, 2011).

This concludes the theoretical background chapter, where research shows that there are a great deal of factors to consider that impact a firm's innovation process from ideation to market launch. Research also shows that integrating or transitioning into a PSS is not an easy feat for manufacturing firms and comes with many challenges, where one determining impact could certainly be the fact that each and every organisation's innovation process is unique or difficult to adapt. With all that taken into consideration, there is still a need to explore this further, as it is not clear what type of challenges there are for manufacturing firms when transitioning and how to even tackle them. Not to mention the need for understanding what criteria are needed for product-service offerings to ensure a promising product for the customers from a manufacturing standpoint, as well as for when the right maturity for launch is.

3. Methodology

This master's thesis is a collaboration between KTH Royal Institute of Technology and Husqvarna. Husqvarna is known to be a world leader in manufacturing premium quality equipment from gardening and forestry to construction. An important note to highlight is that this master's thesis was performed alongside a third colleague, Robert Bunea, who is the author of the master's thesis "*Unwrapping the Customer Value concept and its application within an Industrial Manufacturing Company*".

The method chosen for this thesis was a qualitative research approach which has been proven to be highly effective in order to investigate a field of study thoroughly because of its extensive approach (Hoepfl, 1997). This study also made use of an adaptation of the case study research method by Eisenhardt (1989) aimed to first build a basic understanding from previous cases. To then mainly focus on a single-case study of the firm Husqvarna to build a deeper understanding (Eisenhardt, 1991). This thesis has focused on qualitative evidence during the case study (Yin, 1981). This thesis consists of two main parts, where the first one is the collection of data and the second one is the analysis to bring forth the results. In all chapters, the research questions are covered in the same presented order.

3.1 Data and data collection

The collection of qualitative data, the first main part of this master's thesis, was performed by first constructing a theoretical background of solid blocks for the research questions to stand on. The theoretical background was then followed up by semi-structured interviews and formal meetings with the firm to provide empirical data, including relevant individuals that could share valuable perspectives such as professors, experts within the firm, and other actors related to the firm.

Theoretical background

The theoretical background consists of research articles where search engines such as Google Scholar were used in combination with an evaluation of a journal's quality by taking into account their impact factor. This guided the thesis to strive for well-researched qualitative data to create a solid structure to stand on, for all research questions. The purpose was to understand how and why some actions are already solved or achieved in other organisations to then be utilised for this specific case. The theoretical background played a vital role when analysing the empirical data in order to draw conclusions and evaluations of the results and discussion.

Semi-structured interviews and formal meetings

This study required an in-depth understanding of how current processes are developed and used from an organisation's perspective both internally and externally in order to create a suitable framework for transforming innovation processes. The goal was to collect a great amount of information while maintaining quality. Therefore, semi-structured interviews have been chosen as an information-collective method as it fulfils these requirements. Semi-structured interviews are an effective tool to interact with relevant individuals and grasp an understanding of how and why certain actions are being done, by keeping the subject open while also following a red thread to ensure all relevant data were gathered. While this method allows for a wide variety of choices because of the blend between closed- and open-ended questions, it requires the right application to be effective as it can become labour-intensive and relies on the interviewer's skills (Newcomer, 2015). The work was divided accordingly to achieve the best result possible because of the limited time provided for this project. An interview guide was provided for each interview, see Appendix A, Part 1 for the general interview guide used in this thesis, which was adjusted accordingly to match the informant. Formal meetings were also performed alongside semi-structured interviews, however, in a more open-ended approach. This was an important and relevant method to seek answers beyond the interviewing barriers. Formal meetings were conducted in an environment where the exchange was closer to a discussion rather than that of an interviewer and interviewee. The first out of two main

reasons as to why this method proved to be useful was by lowering the barriers. This allowed informants to share their thoughts more freely and openly, not restricting them to a certain area or topic. The second reason was that formal meetings were also a great leftover technique. For example, questions that could not have been asked due to the time limit during an interview, or questions that were generated after an interview were asked during this occasion instead. This provided additional data to be collected that would have been lost otherwise. All interviews and meetings were recorded and conducted on a digital platform, either Zoom or Microsoft Teams. The full participants of informants can be seen in Table 1 below.

Table 1. *Interviewees from semi-structured interviews and formal meetings.*

Informant	Length	Type
Business Developer A	47 min	Semi-structured
Ideation Director	64 min	Semi-structured
Service Technician	40 min	Semi-structured
Research Director	60 min	Formal Meeting
Digital Product Manager	79 min	Semi-structured
Product Management Director	83 min	Semi-structured
Product Manager A	52 min	Semi-structured
Business Developer B	30 min	Formal Meeting
Product Manager B	53 min	Semi-structured
Product Manager C	51 min	Semi-structured
Professor in Service Design	66 min	Formal Meeting
Professor in Product Innovation Engineering	53 min	Formal Meeting

Observations

Observation is one of many qualitative research methods (Silverman, 2010). It is a good way to discover reasonings and less obvious findings, while also gaining a deeper understanding of what is being observed. There are two types of observations. These are

non-participant observations and participant observations. In a non-participant observation, the observer will observe the situation from a distance, so as to not affect the outcome. Whereas a participant observation includes the observer in the setting that is being observed. For multiple observers, it is advised to take notes separately (Busetto, Wick & Gumbinger, 2020). This is to keep the method objective and increase findings. Observations being an objective method that can be used to see what people are doing through actions rather than described through words should always be combined with, for example, interviews where informants will be in focus, receiving an understanding of how users are feeling and why they performed certain actions (Osvalder, Rose, & Karlsson, 2009). The most suited observation method in this study will be non-participant observations in order not to affect the observed individual and to receive the most amount of data in a shorter amount of time. One 60-minute non-participant observation was conducted and taken into account for this master's thesis. The observation was performed in a store owned by the firm to give an understanding of how end-users experience the full process of using an already established service. The observation was done in the early process of this thesis to give a real-life example and a foundation of how it is already being solved today. This data could then be taken into account, compared and built upon when establishing the finished framework for future product-service offerings.

3.2 Data analysis

The data analysis was the second main part of this thesis. This was performed by first extracting valuable pieces of information and breaking down the gathered data into relevant and significant groups referred to as categories. The importance of conducting an analysis was to provide insights and create an understanding in order to examine and answer the corresponding research questions. The results of the analysis, the findings, could then be reflected upon with empirical and theoretical data in mind, in order to highlight discussion points and to provide a guide to ease the process of transitioning.

Extraction of data and coding

Before the empirical data could be analysed, an extraction and grouping of information, coding, was needed to be performed. The first step was the transcription of all conducted interviews and meetings. This was important, to represent and conceptualise the data before analysing it (Oliver, Serovich & Mason, 2005). For an efficient use of time, an automated transcribing tool was used, built into Microsoft Word. Although the transcription was not entirely perfect, an almost fully understandable version was created which later on was revised manually to ensure that important parts were adequate for use. After transcribing the semi-structured interviews and formal meetings, coding different subjects was the important step before applying the analysing method. The coding was performed by first highlighting codes of any interesting and valuable information said by the informants, which could be directly linked to any of the research questions or just by being compelling. Codes could be anything from a single word to a phrase or a sentence (Saldaña, 2016). The highlighted segments were colour-coded where each colour represented a specific subject. An illustrated example of this can be seen in Figure 2 below. These subjects were then used in the analysis to create categories (Elliott, 2018).



Figure 2. A visualised example of highlighted codes relating to different subjects.

Data structure analysis

The coded data was then analysed to obtain a deeper understanding of the current state of the firm in order to answer the research questions. The data analysis was performed by first arranging the highlighted codes, into groups of the same colour. The same colour implied that there was a common connection between what had been said or expressed verbally. The common connection of codes is referred to as categories. When the categories were established, an iterative process was then initiated to extract additional data from the transcriptions that were missed out or not yet found, distinguishing an emerging pattern. This method helped to link and connect a collection of data that otherwise could have been uncorrelated, see Figure 3 below for a detailed visualisation.

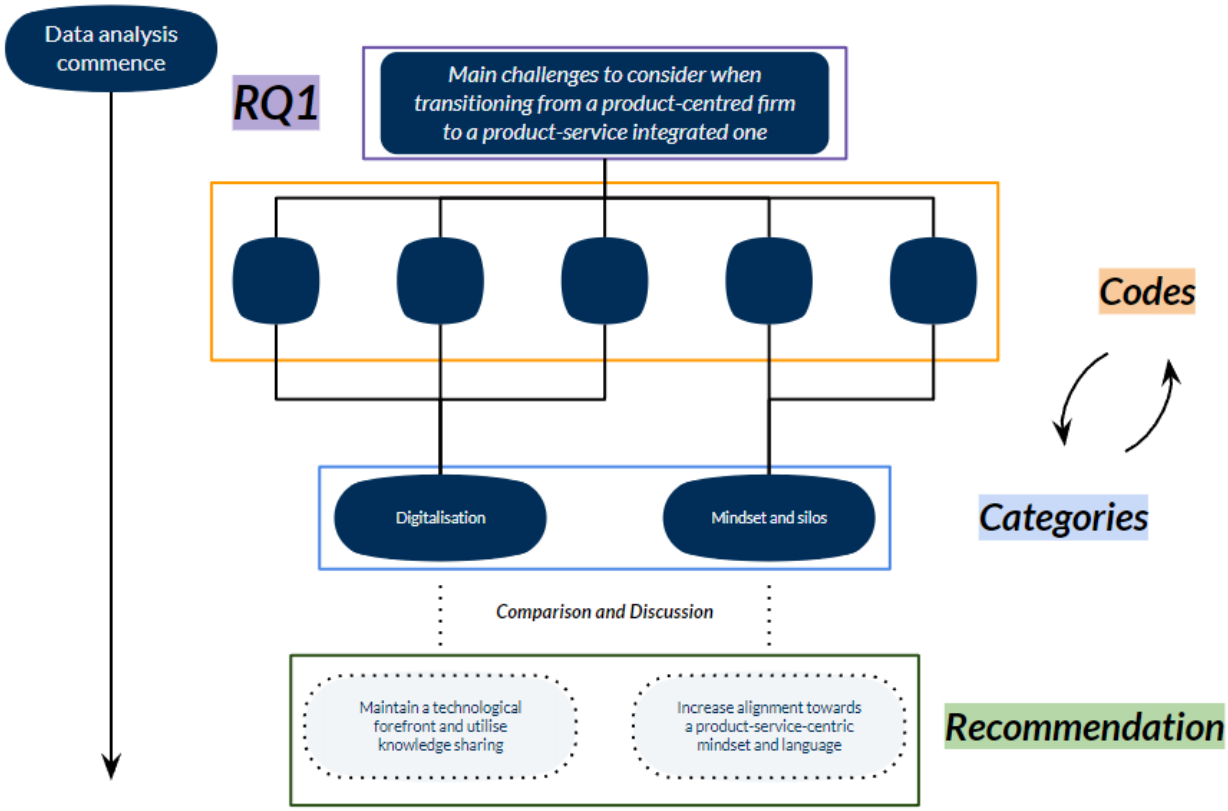


Figure 3. Data analysis method used in this thesis.

The results of the findings from the empirical data were then analysed together with the theoretical background to give further interesting highlights into discussion and also the

resulting proposed framework in order to transition from a product-centred firm to a product-service one.

3.3 Shared informants for two theses

This master's thesis has been conducted alongside another thesis sharing the same informants in semi-structured interviews and formal meetings. This can certainly be seen as a loss of individuality and unique acquired sets of data for each thesis. It can also provide an additional risk of sharing the given interview timeframe, lowering the chances to cover all research questions for both theses. Measures were therefore taken to prevent negative outcomes.

The first prevention was the loss of individuality and uniqueness for each thesis which was prevented by constructing the research questions accordingly. This thesis focuses on a guide taken by firms to ensure a possible transition from a product-centred firm to a more product-service integrated one, while the other thesis covers the customer value in the process. Each theoretical background as well as all interview guides were built separately for both theses. The time was undoubtedly the unknown variable, to give enough time for each thesis during interviews and formal meetings, however, informants were still available for questioning and follow-up meetings after the official interview or meeting which allowed for a full cover of both research areas.

The interview guides were constructed to also cover blind spots for both theses due to the limited timeframe and for increased efficiency throughout the collaboration. This proved to be effective when extracting data, where important sets of data could be found in the additional half of the interview for the respective thesis, which would not have been possible if the focus had only been on one research area. However, citations and data used from interviews and meetings were marked to prevent them from being used twice to ensure individuality and uniqueness, which implies that all citations in both theses are completely unrepeated. This collaboration also ensured an additional depth to be covered

by combining the transition to become more product-service-oriented and customer value.

3.4 Methodological assessment

For a case study research approach, it is important to evaluate the reliability and validity, especially because depending on what type of case study is being conducted, whether it is a single, multiple or cross type case study, results may vary (Voss, Tsikriktsis & Frohlich, 2002). The advantage of performing a single-case study is acknowledged to allow for an even greater depth of understanding of the respective firm in question. However, the highlighted limitations that are present when performing a single-case study are the possibility of biases concerning the misjudgement of representativeness on a single event and the exaggeration of data. There is also a risk of generalisability of the drawn conclusions. This could be mitigated by conducting a multiple-case study, which would arguably provide a wider perspective to allow for more solid results. Also, a larger amount of informants, observations and gathering of data in general, would have contributed to a higher validity of the results as well.

This thesis is solely focused on a single-case approach due to the given timeframe, in order to achieve the most thorough analysis possible. Therefore, a focus on qualitative data was one of the main goals, to allow for the broadest perspective. The interview guides were constructed from the same basis, but tweaked accordingly depending on the informant. This was done to ensure that previous statements expressed by the informants could be validated. The outcomes of this thesis resulted in a guide on a managerial level to aid in the transition towards service-integrated offerings for product-centred firms. Results from this thesis can also be easily replicated, due to the transparency in how the methods have been applied, together with their discussed benefits and potential weaknesses. The findings can also be generalised because of its adaptability and applicability on multiple levels, to the context of other organisations, both within manufacturing industries, but also other industries entering a similar transformation.

4. Empirical data

This chapter will cover the empirical data derived from semi-structured interviews, formal meetings, as well as an observation. All data has been taken out of recorded and transcribed interviews as well as notes taken during non-recorded meetings and observation.

4.1 The current perceived state of the firm

As a large enterprise that maintains its reputation of high-quality premium products, also known for its ideation and creativity, the firm remains a leading competitor in the manufacturing field. Through several recent technological advancements in the last decades and the launch of novel services, it could be argued that the sole focus on the sale of products has shifted into a growing state of transforming into a product-service-focused firm. This piqued an interest to study how employees evaluated and viewed the current state of the firm. It was noticeable in the findings through interviews that their views differed throughout the departments where some informants already could see the firm as a product-service-focused one, while some still could see the transformation as imminent or ongoing.

"We are discussing if there should be add-ons that you could buy more functionality... You have a basic package and then you can add this service as well, however, we're not there yet." - Informant Product Manager A

"PCP product creation process ... A very complete comprehensive process for developing products which takes an idea from definition to product ... And that is now being updated to also include services etcetera and digital components." - Informant Ideation Director

Although the evaluation and point of view of where the firm's current state was in differed, some apparent development of product-service offerings has clearly been shown throughout the years, and a journey from a product-centred firm has already begun, showing great fulfilment in transforming or transitioning towards a product-service-focused firm. Recent projects that were mentioned such as Tools For You, Husqvarna

Fleet Services and NERA, the boundary wire-free robotic lawn mowers are all distinguishable examples of projects with a product-service at the core.

"If we talk about boundary wire freedom that we've had now for a couple of years, that was one very big step for us." - Informant Product Manager A

Changes within the firm regarding the implementation of digital assets on products are being mentioned by all informants. However, the speed of development and adaptation varies between the departments which can also be argued as directly being correlated to how informants also view what state the firm is in. This is partly being described as due to the feasibility of implementing a new feature depending on the respective product's capability, and also the size of the department. There is without a doubt, a core part in need of expansion and satisfaction, a rapidly growing demand when introducing product-service offerings that are being mentioned, and that is digitalisation.

4.2 Digitalisation

Digitalisation is not a new subject that has risen recently over the last few years in the firm as digitalisation was already a core subject more than a decade ago, findings show. Informants describe the importance of always looking ahead in terms of investigating the needs of customers and demand on the market which resulted more than a decade ago, the organisation realised the potential of digitalisation. However, although not being a newly discussed subject, the term digitalisation and its value is steadily increasing across all departments in the organisation in order to provide new product-service offerings for their customers.

"It was decided to start a fully separate organisation for this called Digital Services and Solutions and that was created with the sole purpose of developing digital services." - Informant Product Manager C

Findings show that all departments are somehow involved and aware of the undergoing digital transformation that is taking place, however, albeit in their own enclosures and

boundaries. This change is explained as an important step towards a larger quantity of product-service offerings being launched; where digitalisation more or less is the heart of the process.

"Husqvarna and many other companies are like big oil tankers. It takes a long, long time to change the direction." - Informant Product Management Director

As the firm is a large-moving enterprise, it is to be expected that all departments cannot evolve and adapt at the same rate of speed. However, a rather visible key to access and open up for digitalisation to sprout is the usage of electrical products. Not only is it explained to allow for less complexity when introducing a new feature, it is also essential in order to broaden their product-service possibilities. Some products that are explained to still remain in their petrol state can be seen as a possible hurdle for implementation of digital assets and development. The firm is well aware of this dilemma and measures have been taken for the organisation to not miss out on seizing opportunities. One example of the measures taken are long-term goals that visionise a fully electrical-driven market of chainsaw offerings from the firm within a few decades, globally.

"Then we have our petrol products which typically are not connected… But our electrical products will probably have easier to be connected with Bluetooth, at least to begin with." - Informant Business Developer A

Previous large implementations have already proven to immensely affect the whole product development process chain entirely, from start to finish of a product such as methods, models and tools used. This clearly shows a constant need for transforming and adapting the product development process, now more than ever, especially for the arguable ongoing technological shifts or changes such as the development of AI and machine learning that are taking place today.

"All development processes… How you develop things was quite fundamentally different in all the processes we had compared to doing a digital development where more, agile thinking, agile processes." - Informant Product Manager C

4.3 Transforming the Product Development process

The constant development and adaptation of the product development is entering yet another transformation with the sudden market shifts and technological advancements to name a few. This promotes a need for a sustainable approach as teams' product development processes will constantly evolve, as it already has been, findings show. The first aspect is the rate of speed the processes are changing at, as already described, a large oil tanker is not easily steered swift and agile. Secondly, all departments tackle their problems in the best possible way, which is arguably different for each and every one. Thirdly, the question still remains in how a large organisation can adapt to a changing environment changing even faster than within the firm. One great approach mentioned by some informants was utilising collaborations and expertise outside of the firm. Some reasons mentioned that proved to be beneficial were because of the reduced risks and costs this type of collaboration could provide if established correctly.

"From being more or less only hosted by consultants, and consultant firms to being more and more built up by employees." - Informant Product Manager C

Informants also mention the changes in how lead times have become shorter in the last few decades. This is also evidence of how transformation is always present, as this could be vital to stay competitive and achieve shorter time-to-market which is key in today's field of innovation. Although this has already been performed, a need for an even shorter lead time is being mentioned by some informants.

"The development takes time and the ideas are more than there is time. So I think the biggest frustration within our team is why can't we do this faster." - Informant Product Management Director

When limitations occur within a project, approaches will be switched up, researched and tested if it works. The teams within the firm perform a lot of prototyping and create new and different software when needed. After they have made changes, they go to the

customer to receive feedback, and based on the feedback, it will determine what needs to be fixed. This is a common approach and also an iterative process that is repeated throughout the whole project.

The firm has a few different experimental teams, and within these, there is a lot of pre-development, even though R&D is in the primaries for development. By observing customers and how they are using the products, the team adapts their solutions based on their problems and parts of the offerings that are lacking. Many times the answers can only be found through observations as the customers oftentimes do not give suggestions on what could be done better as they are already quite satisfied with what they have. This information can then be handed over to R&D.

"When we meet customers around the world and we see how they are using products and we see the hurdles and we see the pains that they have and we like to find a solution." - Informant Product Management Director

The firm sees the need for the development of new features, for example, to receive more data from the products and support upcoming offerings. The informants mention that it is much easier to add new features to software compared to hardware, even though the software is newer to the firm. Software is however more uncertain to the firm in how, what part of it, how much and when it will bring profit. However, despite the uncertainties, it is important to release new features early to fight the competition.

"Just being on par, being a premium brand, the expectations in terms of new features and functionalities; You need to be early on. We see and know that our competition is patenting a lot of things, that they're launching new features that they're making." - Informant Business Developer B

Overall the firm sees the need for new resources, however, this comes with both positive and negative consequences. When bringing in new competencies, some see this as a challenge as it takes additional time to teach new employees and familiarise existing staff with new technologies. New employees do not know the established procedure or goals of

the firm and therefore this can cause more mistakes. However, they are much more willing to learn new processes, methods and tools in comparison to senior employees.

4.4 Mindset and silos

A finding regarding the internal agreements and disagreements within the firm is that different departments are mainly working in what is described as silos. This is a very good approach for separate projects where it is explained that each department approaches its cases differently with, for example, innovation processes, methods or tools that are used. However, it is at the same time potentially limiting communication between one another, also not allowing the departments to grow at the same rate of speed, which will be needed for service integration. This shows that different departments of the firm do not necessarily have a fully shared mindset as well as limited collaboration between departments within the firm. Some informants do however mention that in cases where they have ideas that they see would immediately benefit from a collaboration with other departments, they do not hesitate to reach out. The firm also prides itself in its strong collaboration with start-ups and academia.

"We need to agree on who should pay for what, and as long as you stay within a silo… The silo approach is actually really good… The challenge is when you develop services that want to connect different silos. And if those silos exist in different divisions in our case, then the magnitude of the problem becomes even bigger." - Informant Research Director

On the contrary, some informants mention instances of knowledge sharing. Saying that a great deal of individuals in the firm wants to explore and experiment, but the different departments also try to make sure everyone is focusing on the right things, and are on the same path, to prevent waste of resources. Some of the experimental teams have the mission to reach out internally to different parts of the organisation to inform and educate the organisation in terms of new technology developed in the experimental team. Proving that knowledge sharing still happens throughout the organisation.

Investigating the value the firm focuses on, financial value is often mentioned by the informants. However, the informants also point out the importance of customer value to achieve financial value at all. Pushing that customer value actually is the most important driver. Suggesting that there needs to be a balance between financial gains and customer satisfaction. It is also mentioned that anything that does not add value to the end user does not need to be added.

"If we don't add customer value, we are not getting a payback either... It is the customer value that drives innovation and our road map and to make sure that we are relevant. On the front edge of technological development.... if we don't bring customer value, we don't have anything to offer."
- Informant Product Manager A

4.5 Relationships (Customer & Dealership)

The firm has many different suppliers, for example, sensor suppliers and microcontroller unit suppliers. These suppliers also have experimental teams that the firm can take inspiration from and implement the technology in their own way.

"Engaging with the suppliers as these multinational, international companies who also are forming AI labs (experimental teams) on their end, so it is an ecosystem that's sort of starting to take shape now and where in some cases we're getting input and feedback from." - Informant Business Developer A

A lot of interviews and surveys are being conducted to engage with existing and new customer segments and understand their problems and solutions as of today. These answers are then assessed for if they could be solved differently with new technology that is being introduced in the firm.

The firm has dealers in the thousands, globally selling their products. The dealers will have direct contact with the customers, and will have their own lists of what the end users are unhappy about that is then relayed to the firm. This is because there is no way for the firm to have its own staff sell and cater to every end user. However, with the help of digitalisation, they can still have customer relationships, for some product categories.

Notably, it is much easier to collect data from electrical products compared to petrol products. The informants mentioned that the relationship with the customers is not just transaction based, but continues as long as they are a user of the product. These customers should further be served through new updates and support along with the growing amount of other customers. Some of these end users from all over the world act as ambassadors for the products by promoting, using and giving their inputs directly to the firm. These ambassadors can be the first end users to try both software and hardware.

"The relationship doesn't stop when they have bought the product." - Informant Product Manager A

According to the informants, it is always good to test products as soon as possible in practice. There is a quality assurance team that tests the products as end users on the field for first-hand testing, but the first tests usually are set on site.

4.6 Decisions towards a worthy concept

Ideas are identified and acted upon together with stakeholders, experimental teams, data scientists and data engineers. For the experimental teams, the first steps for an idea to go through is if it can be proven as technically possible and that it is concerning an area that they or other teams want to solve. A sufficient number of ideas will be generated as not all ideas will prove to be technically feasible or good enough. While other ideas may bring value, but not be possible with today's technology and the availability of resources. After prioritisation, some will go into smaller pre-studies to be proven. Once this is done the ideas have to be summarised into briefs and go through a core group strategy to see if they are aligned with the mission and direction the firm is heading in. Briefs are seldom denied as those submitting the ideas know the vision of the firm very well.

For the experimental teams, once an idea has been proven, but is not seen as viable as of that moment, it will be worked around rather than killed. There is constant iteration until a concept can be accepted for the next step. It usually takes 12 to 15 weeks of iterations before it can be verified with tests, research and prototyping. Data is then needed to see

if it is feasible to go through with. After that, the concept needs to be taken through a number of steps before it is ready for scaling. If there is not enough data, if it is too difficult to generate and if no one is willing to pay for the data the concept may need to be denied unless that can change. Luckily, many of the rejected ideas turn into learnings for future projects.

"We also learn when we decide not to take them... What's not possible today might be possible tomorrow, then we know we can pick up the idea." - Informant Ideation Director

Each project has a separate document specifically written, by the project owners, for that project. This document includes all the market user requirements, mentioning everything the product is supposed to achieve when it reaches the end customer. So if there are possible ways of cutting corners to save time and money, certain things mentioned in the document cannot be touched, as the product then would lose its purpose and value to the end users.

There are still concrete constraints that are mentioned for concepts to be realised. Mainly if the hardware software requirements can be met. The ideas for digital products need to be prioritised based on if they can deliver on consumer values while being long-term care solutions with high quality, done in an intelligent way that brings something new and extra, compared to competitors. The informants say that if the customer values are being met, then they will also see business growth. Some projects will be chosen if they have a high chance of success in bringing in a profit, however, projects will not necessarily only be chosen based on these criteria.

"We believe that if we launch this and give it to the consumer. They will feel more in control, they will have more time, and they will feel smarter... we believe we'll see that outcome and then we will get our business outcome." - Informant Product Manager B

End users are mostly involved in the discovery of new issues and solutions, but not as much in the rest of the development process. Instead, there are system testers and quality

assurance teams that act and test the products as end users to create an understanding of the end user's perspective.

"We don't quite involve end users in the whole development phase. It's more in the beginning and then when they [the products] hit the market." - Informant Product Manager A

4.7 The unknowns of launching a new type of offering

Findings show that the launch of a product is very complex and unique for each and every product. Informants describe that the core features are the first and foremost important goal to reach while any additional features or parts will always be secondary, in other words, the 'a car cannot drive without the four wheels principle'. Some informants also explain that an important set of criteria has to be met before even considering a launch.

It is understood within the firm that services can bring value to the consumers, therefore they are investing in the experimental teams, to look into and experiment with digitalisation and services, with support from product management teams. The goal for the experimental teams is to verify the value the services can bring before they can be proven to the whole firm.

There is an aim towards being the first on the market, but when the firm is not, they do not see it fully as a negative. Instead by already having competitor's products on the market they know the market price, and can therefore estimate the profit based on the development cost, the market price and the brand quality. This can however not be done with as much certainty when it comes to services as with products due to being a fledgling in that area. There is also the case of the unknown when new ideas have not yet reached the market. In this case, the firm takes its learning experience from both approved ideas, as well as rejected ideas that have been proven impossible. The informants say that if a project is proven impossible on their end, they can rest assured that it will also be impossible for the competitor, therefore it is one less thing to worry about. The firm is very aware of its competitors and understands that to stay as a world leader in its market,

it would need to be bold and brave regarding investing and taking chances, taking risks, which is also a reason why the firm is now investing in service integration.

5. Discussion and analysis

The discussion and analysis in this thesis create a foundation in order to get a better understanding of the transformation and the process for manufacturing firms to become more service-integrated. In this chapter all data collected is analysed to discover the skills firms need to overcome obstacles related to service integration, choosing the best product-service offerings to advance in the innovation process, and understanding when to launch such offerings. In other words, answering the research questions of this thesis. This will all be compared to the case studied and discussions on improvements will be provided.

5.1 Challenges to consider during servitisation

The literature mentioned multiple challenges to consider when transitioning to service-integrated products. Some vital points had overlapping challenges and other points included multiple challenges. It could be concluded that the following challenges, taken out from the literature, can be regarded as what is required of manufacturing firms to consider when transitioning towards service integration. In this master's thesis, these challenges have been divided into three categories:

1. State of mind within the organisation.
    - A shared language within the organisation for efficient knowledge sharing and in-house collaboration (Vasantha et al., 2011; López-Nicolás & Meroño-Cerdán, 2011; Baines, Lightfoot, & Kay, 2009; Martinez et al., 2010).
    - A shared mindset for service, based on transparency, is to be aligned within the whole firm. With this, the firm should develop its product-services as a whole and not as separate parts (Baines, Lightfoot, & Kay, 2009; Isaksson, Larsson & Öhrwall Rönnbäck, 2009; Story et al., 2017; Martinez et al., 2010).
    - Integrating a product-service culture and becoming more customer-centric. In turn, going from transaction-based to relationship-based value (Story et

al., 2017; Schein, 1990; Vargo, Maglio & Akaka, 2008; Martinez et al., 2010).

2. Acquirement of new resources.
    - Create new systems, processes and tools adapted to services (Wallin, Parida & Isaksson, 2015; Martinez et al., 2010).
    - Acquire new capabilities and competencies focused on customers and services (Wallin, Parida & Isaksson, 2015; Martinez et al., 2010; Isaksson, Larsson, & Öhrwall Rönnbäck, 2009).
3. Relationships outside of the organisation.
    - Increase the relationships and communication with suppliers (Martinez et al., 2010).
    - Co-creation with the customer throughout the whole project, as well as an increased relationship and communication with the customer for a higher understanding of customer needs (Isaksson, Larsson, & Öhrwall Rönnbäck, 2009; Mahr, Lievens & Blazevic, 2013; Pitta & Franzak, 1996; Thomke & Von Hippel, 2002).
    - Retaining a sustainable transition (Beuren, Gomes Ferreira & Cauchick Miguel, 2013).

The Empirical data shows an area of potential for growth regarding internal collaboration and knowledge sharing, due to the current work process based on silos- This is currently being combated as new technologies are being introduced. Nonetheless, the firm is not hesitant regarding external collaboration, as it shows strong collaboration with both start-ups and academia. The firm is still split regarding its position on the path towards a product-service-focused firm. Therefore also not having a shared mindset for service and alignment within the firm, and mainly work in silos. Looking at category one which was derived from the literature. It is an area that the firm may need to work on to achieve full transitioning for service-integrated products.

In contrast to the literature that points out the need for a full transformation from transaction-based value towards relationship-based value (Vargo, Maglio & Akaka,

2008), the informants mention the value of keeping a relationship with the customer even after a transaction has been done. This brings forth the importance of keeping a balance between transactional value and value based on customer relations.

The empirical results showed that there is a need for new features and therefore also a need to develop them, as a constant adaptation towards what is required to meet new standards and needs. The challenge is not in whether or not the firm can acquire new resources and tools, but rather the lack of experience of using them. Digitalisation and services are rather new to the firm and therefore unpredictable of what they will result in. This is where the firm can find the strength in new competencies and personnel that is more open towards uncertainties and change. Moreover, this is an area where their external collaboration can shine through, as that way they can then gain access to knowledge they have yet not experienced. Therefore, the firm is on the right path as well as having a steady future regarding category two.

The firm has strong relationships with both suppliers and dealerships, for communication and collaboration. There is a valued relationship with the customer, however, this relationship does not regard the full development process of new offerings. There is constant feedback and data coming in regarding customer needs and problems, giving insight into what needs to be developed and improved next. However, there are limitations to the co-creation and collaboration with customers in the creation of offerings throughout the whole development process, which would be the only drawback regarding category three. In addition, the firm makes sure to promote a sustainable approach during the transformative process, not only creating a stronger trust with customers who would rather choose a sustainable offering but also positively strengthening the firm's relationship with the environment.

5.2 Criteria of a successful product-service offering

As for criteria required for an idea or concept to be good enough for continuation, it can be concluded through recurrence in the literature that certain criteria are more critical

than others (Hart, et al., 2003; Balachandra, 1984; Cooper & Brentani, 1984; Frederiksen & Knudsen, 2017).

- Technical means to fulfil requirements of the creation of the offering.
- There is access to materials and physical resources required.
- Potential for profitability, high sales, as well as company growth.
- Having a chance to succeed in the market, optimally outperforming competitors.
- Offerings can be created with high quality and performance.
- It is a novel and unique offering.
- There is communication and collaboration between the firm, dealerships as well as customers.

When specifically analysing services to the above-mentioned criteria, their main similarity is customer involvement. However, for services, it is emphasised that the customer is required to be involved throughout the whole development process. Early on regarding service offerings it is of most importance that the offering can be of high quality, whereas later on profitability is prioritised (Dörner, Gassmann & Gebauer, 2011).

The firm makes sure to choose projects that have the technical means and resources to create the desired offerings, and in cases where there are limitations or unexpected hindrances, the teams will work around the issues, through iterations, until the offering can be released.

Most concepts are chosen based on the chance of success and profit. However, not all projects are chosen based on profitability, but instead on how closely they align with the vision of the firm and can lead to growth and customer satisfaction. The firm prides itself as a manufacturer of premium products of high quality and lifetime, added to this is the long-term care and relationship the firm has with the consumer. Being a leader in the market, the firm always strives towards being early on regarding the release of new and novel features and functionalities.

The findings show that new solutions are many times based on problems derived from users. This can be done directly through the product ambassadors, simulated by system testers and quality assurance teams, or passed on from dealers that have had direct contact with regular end users.

The current decision-making process done at the firm takes into account the majority of criteria found in the literature and has therefore a high chance to create successful product-service offerings.

5.3 Launch of product-service offerings

Patterns show that the relative size of a firm normally determines when projects are being launched relative to the market. To launch a successful project it is important that certain elements are being met by the firm (Bayus, Jain, & Rao, 1997; Tholkea, Hultinkaa & Robbenbb, 2001).

- Having an understanding of the value services can bring to the firm (Chester Goduscheit, & Faullant, 2018; Sklyar et al., 2019).
- Having an understanding of the competitors (Bayus, Jain, & Rao, 1997; Grawe, Chen & Daugherty, 2009).
- Having an understanding of potential sacrifices and the rewards they can lead to (Bayus, Jain, & Rao, 1997; Ettlie & Rosenthal, 2012).
- Having managerial incentives towards the growth of the services (Dörner, Gassmann, & Gebauer, 2011; Ettlie & Rosenthal, 2012; Miles, 2010).
- Having a direct customer relationship and feedback loop (Garud, Jain & Tuertscher, 2008; Lehmann & Recker, 2022; Rapaccini et al., 2013).
- Having an MVP for initial launch (Ries, 2011; Maurya, 2012).

Direct and constant customer feedback, relations, and collaborations has been mentioned repeatedly throughout this master's thesis. However, studying a manufacturing firm it is natural that customer co-creation of products throughout the whole process is limited. This is because there are no appropriate or appealing ways for the customer to take part

of the development process of physical goods, outside of the beginning, where they can inspire what product should be created next, and at the end, when it reaches sales. It can be believed that like traditional products, a product-service offering would also need to be fully complete before launch, falling into the same issues with limited customer collaboration. However, by keeping the process and the end results as one, the customer can still be involved in the development process while also experiencing a functioning product (Garud, Jain & Tuertscher, 2008).

For a manufacturing firm early on in transitioning towards increased product-service offerings, the firm can first release the MVP to a limited pilot test group before opening it to the public (Rapaccini et al., 2013; Maurya, 2012). These can be the product ambassadors or other test groups available to the firm. By then officially launching the product-service offering through a soft launch (Couldwell, 2019), customer feedback can be directly taken back and used for further updates of the offering (Garud, Jain & Tuertscher, 2008; Lehmann & Recker, 2022).

The firm already sees the value in services, both within the experimental teams and management. They are therefore prepared to take the chances and go into the unknown, which currently is a concept that needs to be validated and verified before it can be scaled up.

There is also a strong understanding of the competitors and knows how to position themselves and adapt accordingly. As well as an already established feedback loop at the early stages of development, that just needs to be implemented on a larger scale throughout the whole process. However, as product-service systems are still new to the firm, soft launch and MVP have still not been done, as the firm previously only realised fully complete products. Because the firm already has its product ambassadors as an established group of end customers, pilot testing as well as soft launch should not be difficult to implement, together with the concept of MVP.

6. A practical framework towards product- service offerings

This chapter is built upon a data collection that was dissected in the form of a data analysis and discussion covering the theoretical background and empirical data. This gave an understanding of gaps within the present research in this field of study. The proposed framework in this chapter is a completely original creation with our own recommendations based on the discussion and analysis of the gathered data. The purpose of the framework is to aid Husqvarna's journey towards transitioning to becoming more prominent in the product-service offering field; by answering all research questions to the best of our ability. The framework is also built on an elevated level of acknowledging the important factor of customer value, which is why it is highly recommended to see the other half covered by Bunea (2023) to grasp the full picture of this collaboration. See Figure 4, for a complete guide to transition towards successful product-service offerings and each sub-chapter breaking it down in detail.

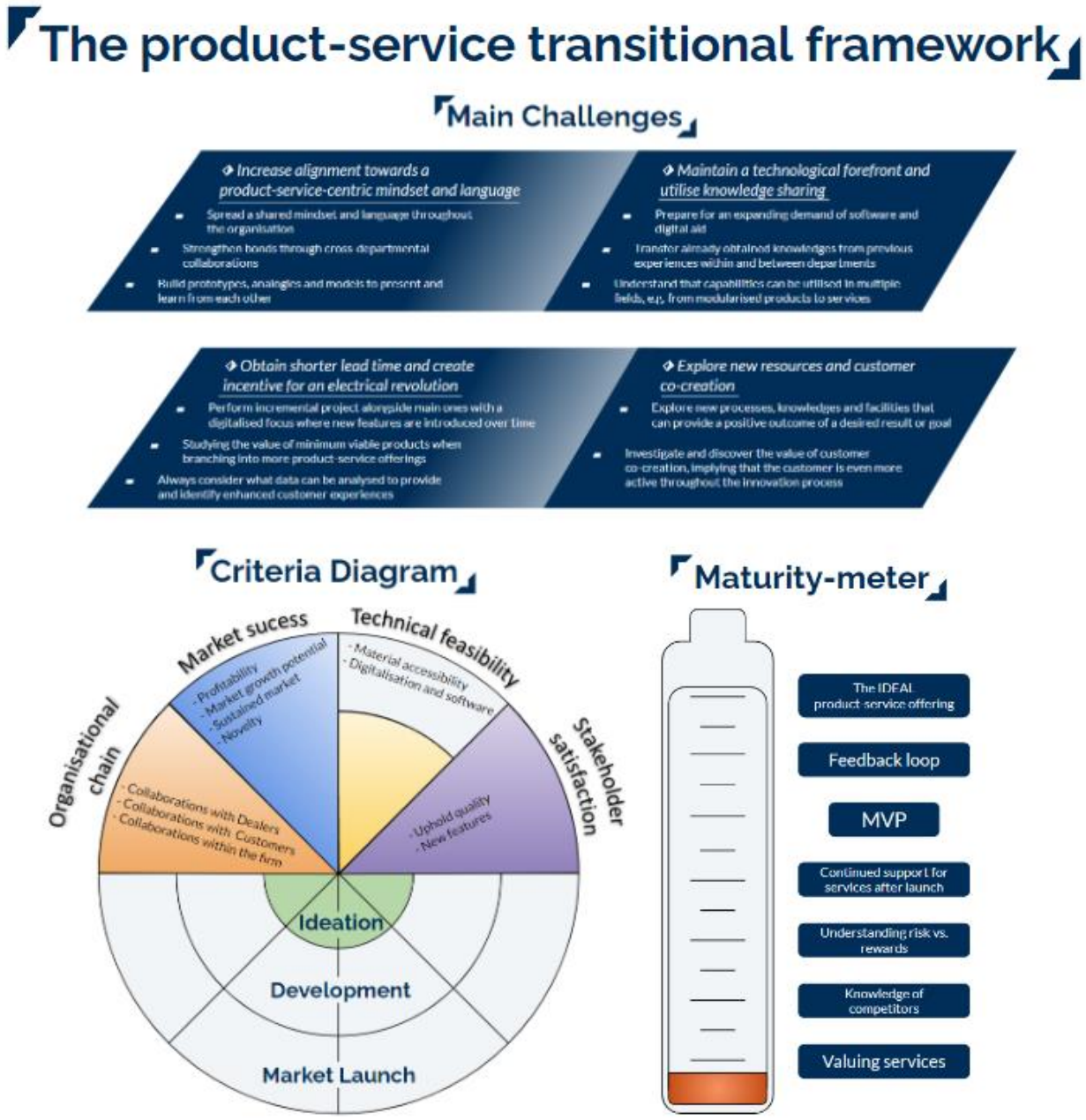


Figure 4. The product-service transitional framework

6.1 The main challenges to consider when transitioning from a product-centred firm to a product-service integrated one

These main challenges are to be considered in order to achieve a great success rate in a product-service integrated firm. See Figure 5 for a summarised version, and a detailed description of each step below.

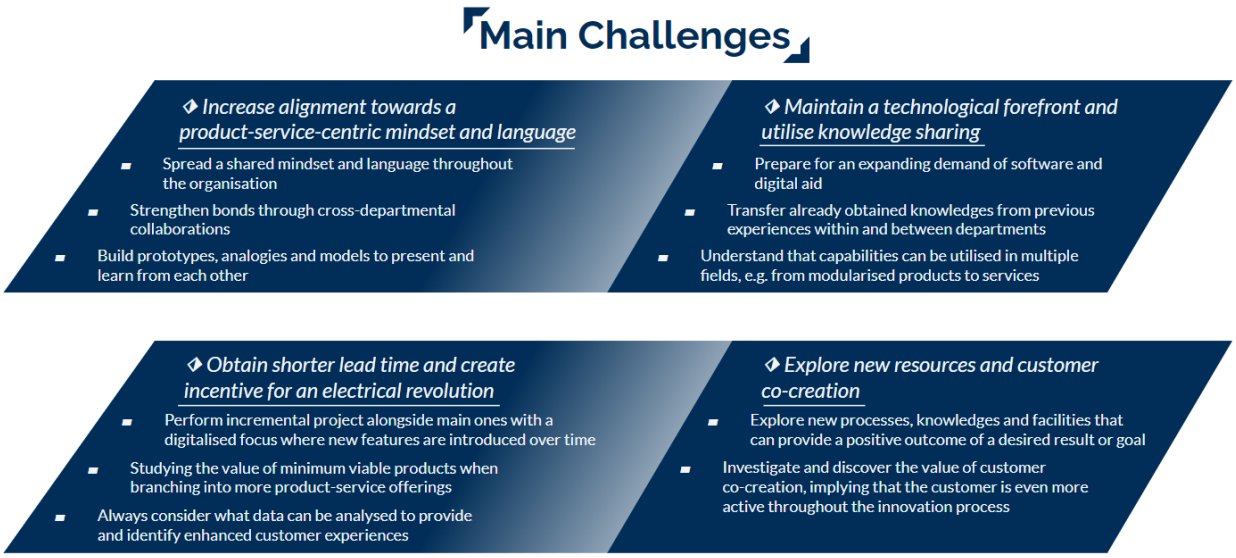


Figure 5. The main challenges

- *Increase alignment towards a product-service-centric mindset and language.*

A first step towards a service-integrated firm would be an increased alignment throughout all departments towards a product-service-centric mindset and language. Right now the firm shows a potential for growth in their in-house knowledge sharing. However, even if departments were to stay as silos; if said silos build prototypes, analogies and models together for visual presentation, the understanding between silos can increase. Meaning that if the firm keeps regular visual transparency between departments for alignment, they can both create a better understanding between departments and also for spreading a product-service culture throughout the firm.

- *Maintain a technological forefront and utilise knowledge sharing.*

The need for digitalisation is constantly rising throughout the whole organisation. A digital transformation within is needed to satisfy the increasing demands from all departments. This includes an expansion of the IT department, who in turn should have an understanding of the combination of digitalisation and innovation. Understand the power and value of transferring already obtained knowledge between and within departments as well, for example, how modularisation can be applied for digital services from previous projects. Be creative and investigate how different capabilities can apply to multiple fields.

- *Obtain shorter lead time and create incentive for an electrical revolution.*

So far, the process has already been developed and changed towards shorter lead times. This is essential to obtain high speed to market or time-to-market. The request to maintain a competitive front is to achieve even shorter lead times. The first main point should be in combination with the previous recommendation, to expand the capacity of technological assistance while creating incentive for long-term digitalised projects to grow alongside the main ones. These projects should be focused on constant development and product-service-focused, introducing more features over time. This should prove to shorten lead times significantly, while studying the effect of MVP. The second main point is pushing for an electrical revolution. The faster a transition of electrical products is made, the faster new digitalised product-service offerings can be achieved. This can also open up a new pathway for new data to be analysed in order to for example, identify customer behaviours and potential predictions of the future outcome.

- *Explore new resources and customer co-creation.*

The firm should take the opportunity to explore new resources focused on product-service offerings, such as processes, knowledge, and facilities, when the possibility appears, while ensuring a sustainable implementation and transformation. This will prove important to reduce risk for negative outcomes such as cannibalism of other products for example,

while also allowing future projects that may rely on these resources can be implemented. Furthermore, to become a product-service integrated firm, they would need to bring more attention towards customer co-creation. Implying that the customer should be involved more throughout the whole development process for constant feedback. This however may only be possible for digitalised offerings that can be released and reviewed by the customer early on.

6.2 The criteria to proceed in a product-service-oriented innovation process

The firm in this study is a leader in their market area and has therefore already a very innovative development process which is close to what is required for a product-service integrated firm. A visual criteria diagram that can be used to determine what the current capabilities are and should aim to be present within the firm. Figure 6 below can be used as an inspiration on how to adapt the innovation process accordingly. Each capability is gradually expanding and needs to remain incorporated towards market launch, eventually reaching a desired state. See Figure 6, for inspiration.

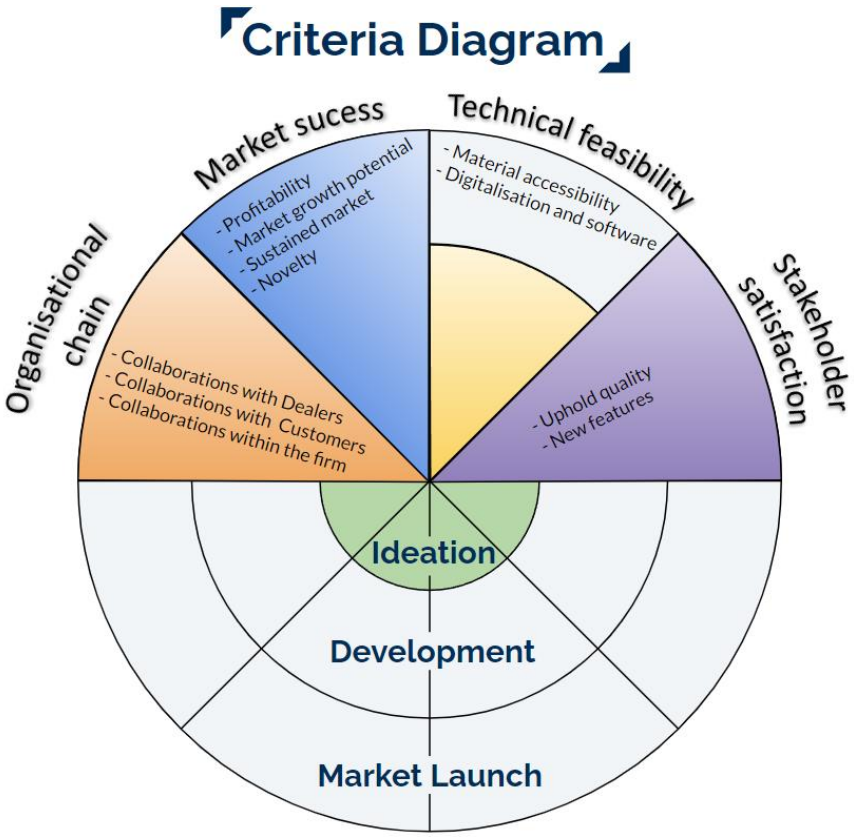


Figure 6. The Criteria Diagram

- *Create unique market user requirements and adapt methods, processes and criteria.*

By having a unique market user requirement document for each new project, the firm can make sure that the customer values are always met, no matter the inevitable changes that happen throughout the long development process. The firm only chooses projects that they see as feasible based on the resources available or if they can be obtained, novelty, as well as its chance for success and profitability. That way, it ensures high quality and performance. There is also a high probability that new methods, processes and criteria will be required to increase product-service offerings. At the same time ensuring a path to allow growth. In order to enter a new unknown region, collaborations and other important third-party actors can lower both risks and costs which can later be turned into the firm's own strength over time.

- *The Idea Bank and cross-departmental collaborations.*

The firms should create The Idea Bank specifically for product-service ideas. This will promote incentives for creating the right innovation environment as well as connect different departments, not on a whim but rather on planned occasions. These planned occasions should include joint meetings, workshops, cross-departmental discussions and collaborations to learn from each other and overflow the idea bank. Alternatively creating a specific team that will focus on closing the gap between the isolated silos and scouring the idea bank to bring up potentials and build upon them.

6.3 The right maturity and relevance to launch new product-service offerings

For a successful product-service market launch, these findings show a higher probability of success if fulfilled. In order to determine a sufficiently matured product-service offering, a measure of how ready a launch is can be seen in Figure 7, which can be used to create an understanding of how ready a product-service offering is for launch, and therefore determine a suitable market launch.

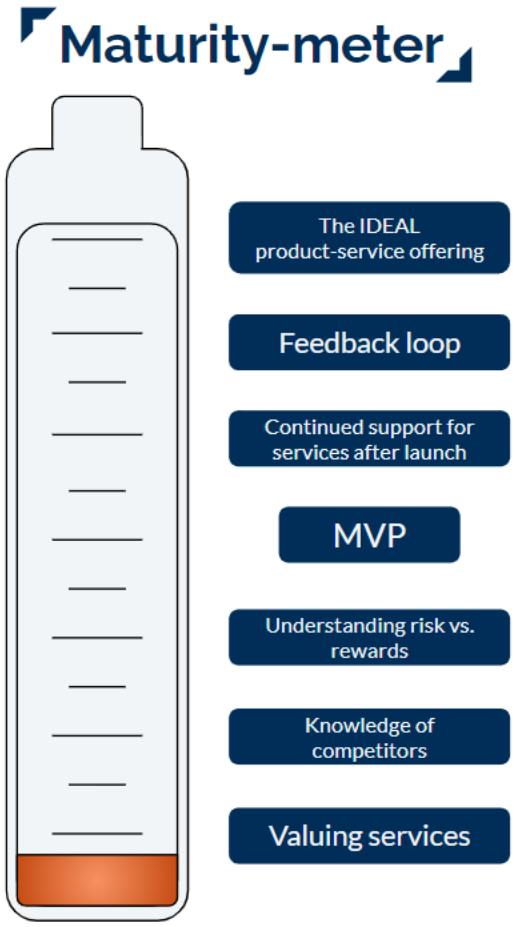


Figure 7. The Maturity-Meter

- *Always have a clear, definitive strategy and goal.*

To even consider launching a product, it always has to be a start and a beginning. No matter what offering, there must always be a definitive strategy and goal involved. Answering important questions such as 'Is this even possible' or 'Will this generate value that we can benefit from as a firm'. Important factors are always to be considered, risks will always be involved and not being willing to take risks is not always good. However, do not just jump on the train because of the majority, there should always be a reason or multiple reasons behind every action taken. Learn from competitors and consumers, even look in similar fields that have been successful. For example, software development, application development and games development are some that have tons of experience in that field.

- *Establish the core blocks for a product-service launch.*

The firm has a strong interest in going towards services, while knowing the possible risks they can face due to their weaknesses in the field, they are already steps ahead towards being able to have a successful product-service launch. However, the core blocks are currently incomplete, therefore, building and completing these should be a priority. These core blocks are the MVP and the constant feedback loop throughout the whole development process. The firm currently has all the resources needed to complete these blocks and is therefore well-equipped for a successful product-service launch in the not-too-distant future.

7. Conclusions, future works and limitations

The road for manufacturing firms to become more service-integrated is not clear, however after discussion, analysis and a thorough generating of a practical framework, this thesis could provide a basis for firms on a managerial level to get a greater understanding of the transformation and process. This will give firms necessary tools needed to more confidently go through challenges when becoming service-integrated, selecting the most successful product-service offerings to proceed into the innovation process, as well as knowing when to launch said offerings.

With that said, in this thesis, it has been discovered that for a manufacturing firm to transition towards incorporating services in their offerings it is important for them to take certain aspects into consideration. In this thesis, three categories have been highlighted as to what is most important to consider and implement when undergoing the shift. The first category covers the state of mind within the organisation, stating that by having an aligned way of thinking regarding services, good collaboration, understanding, and quality become easier to achieve. The second category states that in becoming more service-integrated, new resources focused on services need to be acquired, in order to adapt towards servitisation. The third category discovered in this thesis covers the external relationships with the firm, in which collaboration and communication with customers and suppliers are of utmost importance when undergoing the change from a product-centred manufacturing firm to a service-integrated one. Through examining these results together with the discoveries from the empirical data, it has in this thesis therefore been concluded that the main challenges when transitioning towards service integration are as follows.

- To increase alignment within the organisation towards a product-service-centric mindset and language.
- To maintain a technological forefront and utilise knowledge sharing within the organisation.
- To obtain shorter lead times and create incentives towards an electrical conversion.

- To explore new resources and increase customer co-creation.

Once these four points have been prepared for it can be assumed that the firm is ready to create service-integrated offerings.

This should still be carried on and considered once the transition has been executed, and is therefore still also important during the concept screenings. In this thesis it has been concluded that for an offering to be successful, certain criterias must be met and taken into consideration during developing and supporting. These criteria covers the following:

- The collaborations externally and internally with the organisation.
- The chance for market success, regarding brand growth on the market, profitability and competition.
- The technical feasibility of creating the offering, regarding accessibility to resources and availability to relevant competencies.
- These above points while upholding quality and allowing for new features to be presented.

When the right concept has been selected it can be developed and eventually launched. A product-service offering does not need to be fully completed when released and the margin for when it should be released therefore becomes unclear. In this thesis, it is concluded that for a successful launch of a product-service offering, all the previous points stated for adapting the firm for servitisation as well as the criteria for concept screening of product-service offerings need to still be in place and met. In addition, it is important that the following points are fulfilled before the offering is launched.

- There needs to be an understanding of what value can be gained from the service.
- The competitors positioning and offerings on the market need to be understood.
- Risks need to be understood and weighed against the rewards.
- Most importantly, an MVP must have been developed.
- It should be ensured that managerial support will continue even after launch.
- And lastly there is a customer feedback loop ready to be utilised.

When these have all been run through, and the aspects for servitisation are considered and established, the process for selecting and launching a product-service offering can start over.

When it comes to the right time to launch a product-service offering, there can be found literature discussing points that must be met before the release of an offering. There is also literature that encourages incompleteness and continuous development of the offering after launch. However, there is little to no literature found discussing the level of maturity as well as what functions would be fine to leave out of the initial launch. This would have been useful to this thesis, and would also be interesting for future works.

Some last remarks concerning the fact that the results are based on a single-case study of a large enterprise, further analysis of similar firms would arguably provide a wider and deeper perspective affecting the results. A larger sample of informants, observations and gathering of additional data would also be vital to contribute for a better outcome. However, it was also required to focus on a single firm to obtain the maximum amount of insight in this case within the set amount of given timeframe to provide guidance for a suitable transition. In conclusion, although the study is solely performed on a large enterprise, the result of this thesis can still be relevant for smaller-sized firms as well, as the framework laid out in this thesis is adaptable and generally applicable at all levels.

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Appendix A: Interview guide

Informant(s):
Date:
Time:
Department:
Format:

❖ **Introduction**

- Tell us about yourself, who is [insert name] and what is your role at Husqvarna?

PART 1 - Innovation Process

- Can you briefly explain how the current product development process works?
    - What about the development process for product-service systems?
    - Any examples?
- How do you work with the customers? Are they involved?
    - Which part of the product development process, the early, mid or end?
    - How do you receive feedback?
        - And how do you use it?
- When developing a product, can you briefly explain how and what parts of departments work together within Husqvarna? Outside Husqvarna?
    - Examples?

- Currently, you have some interesting projects such as ***Husqvarna Fleet Services*** and ***Tools for you***. Have you been a part of these developments or is knowledgeable enough to give us any information about how their Product development processes look/looked like?

    - If not, could we get in contact with them?

- Do you have any other experiences closely related to this subject, concerning any product and service offerings? (it does not have to be a finished product)

- Do you currently have any ideas of clear issues that can be run into when implementing product service systems in Husqvarna?
    - Example: Are there any departments that could be sensitive to the change related to the transformation?

- What does the decision-making process look like? Do you use a method?
    - Does it change depending on what "phase" or "stage" the product is currently in?
    - Is it any different to a product-service offering process? Do you have any examples from projects you were part of?
        - If not, could we get in contact with them?

- Do you weigh a product on criteria? [Specific Question]
    - Which criteria?

- How do you decide if you need to kill ideas?
    - Early on as well as later on?

- Do you have any other experiences closely related to this subject, concerning any product and service offerings? (it does not have to be a finished product)

- How do you determine when a product is ready for launch? Do you have any examples?

- How does that change to services? (Fleet Services or Tools for you) Or other projects you have been a part of?

PART 2 - Customer Value

- How do ideas come in to be considered for development?
    - How are these ideas presented and what is being looked after?
        - Is customer value an important factor that you look at?

- How would you define the concept of customer value from your work within product development projects at the company?

    - Do you think that this definition is consistent throughout the department(s) and the company? Why/Why not?
        - Who is responsible for this?
        - Which departments or individuals are investigating and have knowledge of this?
            - Where is this information important for development?

    - How do you work with customer value throughout the development of a project?
        - Why is it important for you to take customer value into account during development?
        - Examples?

- Do you often see misinterpretations regarding customer value?
    - When does this typically happen?
    - In your opinion, what is the cause of such misinterpretations?
    - Can you think of an example of such a situation(s)?
        - Do such misinterpretations happen within or between divisions or departments?
        - How was it solved?

- How has your view of customer value changed over time?
    - What has made your view change?
        - How has this change influenced development?
        - Can you exemplify?
            - One past example with past view of customer value?
            - One recent example with current/recent view of customer value?

- Is there any difference when developing physical products or digital products in terms of customer value?
    - What do you think creates a difference between these different products?
        - Examples?
    - How does this change when developing something that has both physical and digital components?
        - Examples?

- How do you estimate value within a project? Examples?
    - Do the ways of estimating vary between digital and physical products?
        - In what ways do they vary?
        - Examples?
    - How is this estimation done for a project that has both product and service elements? (For example Husqvarna Fleet Services)

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